

Vishal Kumar Rai

Economist | Quantitative & Structural Economic Modeling
Ph.D. in Economics (Thesis Submitted), Indian Statistical Institute, Delhi
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Professional Summary

Economist specializing in how individuals make retirement, health, consumption, and asset decisions under uncertainty. I build and calibrate structural life-cycle models of longevity, morbidity, and occupational health risk to study retirement behavior, life-cycle consumption and saving, health expenditure, and asset accumulation and decumulation. My work combines dynamic optimization, microdata calibration, and numerical simulation using Julia.

Research Areas

- Structural Life-Cycle Modeling
- Retirement, Consumption & Saving
- Health, Longevity & Mortality Risk
- Occupational, Environmental & Epidemiological Risk
- Dynamic Optimization & Simulation
- Quantitative Macroeconomics

Skills & Methods

Modeling: Structural Life-Cycle Modeling, Dynamic Programming, Stochastic Optimization, Calibration, Policy Simulation

Data: Large-scale Household Surveys (MEPS, CPS, PLFS, LASI), Mortality & Demographic Data (SRS, CDC, NOMS)

Computing: Julia (Model Development, Numerical Optimization, Package Development)

Education

Ph.D. Economics ISI Delhi (Thesis Submitted)	2019–2026
M.A. Economics Delhi School of Economics	2016–2019
B.A.(Hons.)Business Economics University of Delhi	2012–2015

Open-Source Contribution

[SwitchingControl.jl](#)

Registered Julia package for regime-switching dynamic optimization problems, developed for solving the structural life-cycle models used in my research.

Research Experience

Health, Endogenous Retirement and Demand for Annuity

Ongoing

Funded by the Department of Science and Technology (DST), Government of India

Developed, calibrated, and numerically implemented a structural life-cycle model of health, retirement, and annuity demand in Julia.

Constructed occupation-specific wage, mortality, survival, and frailty profiles using PLFS, LASI, SRS, CDC, and NOMS datasets.

Conducted policy simulations to evaluate behavioral and welfare responses under alternative pension and tax regimes.

Climate, Pollution and Endogenous Retirement

Developed a continuous-time structural life-cycle model linking pollution exposure, health, labor supply, and retirement timing.

Calibrated the model using microdata and performed numerical policy simulations in Julia.

Endogenous Retirement under a Pandemic

Job Market Paper

Built a stochastic dynamic model to study retirement decisions under pandemic mortality risk.

Implemented numerical solution methods and simulated outcomes under alternative epidemiological scenarios.

Occupational Variation in Work Strain

Developed and calibrated a structural life-cycle model with occupation-specific work strain and health risk.

Used analytical and numerical methods to explain observed occupational differences in retirement timing while replicating empirical life-cycle patterns of health expenditure and asset accumulation/decumulation.

Conference Presentations

Endogenous Retirement under a Pandemic

June 2026

Presented at the CDE–IEDS International Conference on Economic Development, Demographic Transition, and Environmental Sustainability, Delhi School of Economics.